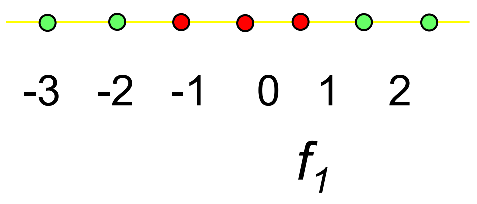
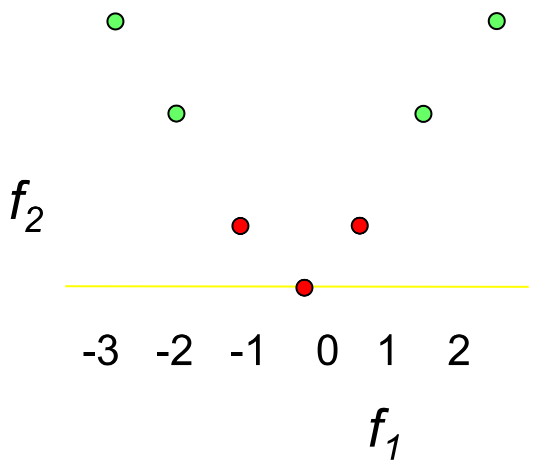
Simple Quadric Example

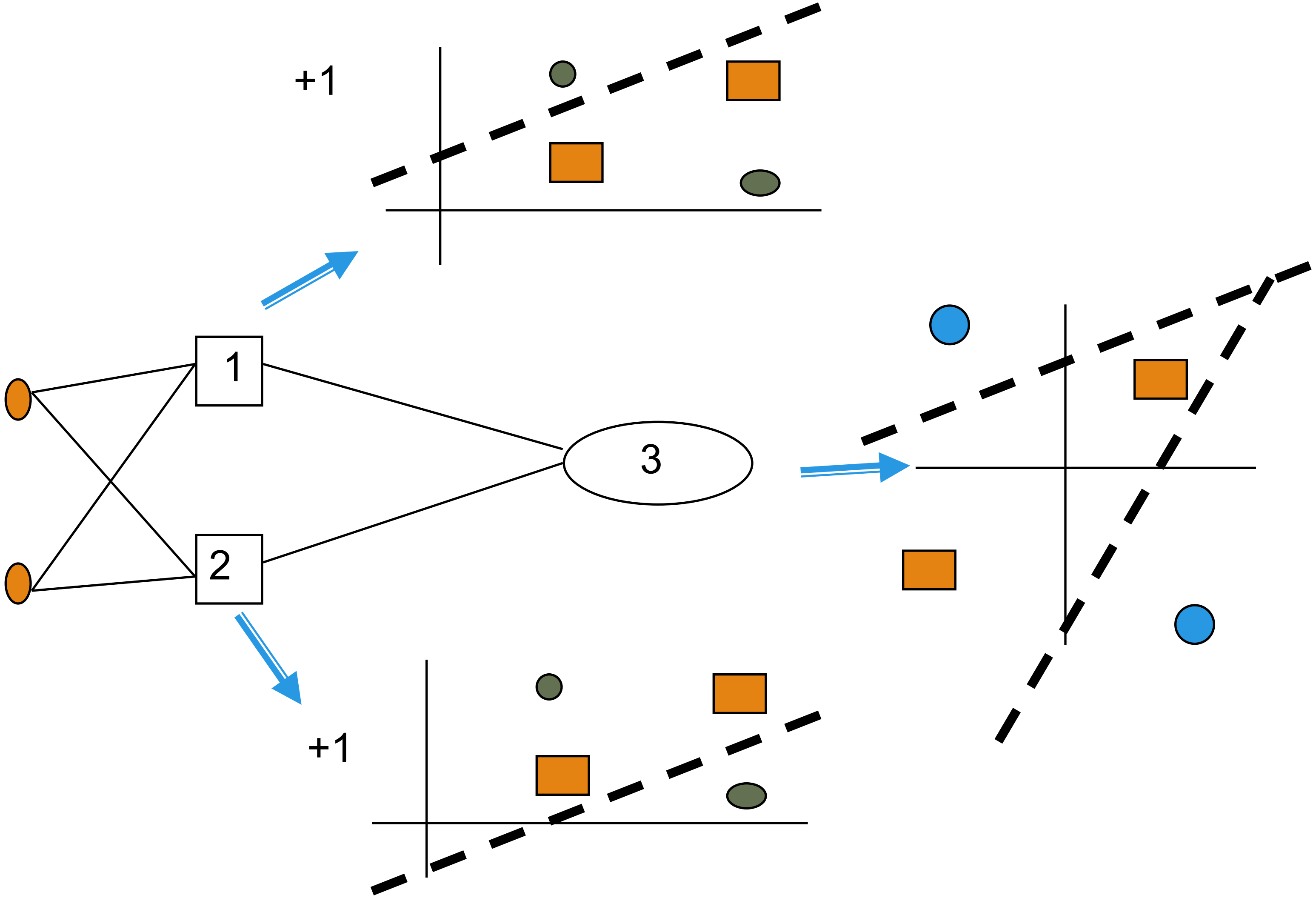




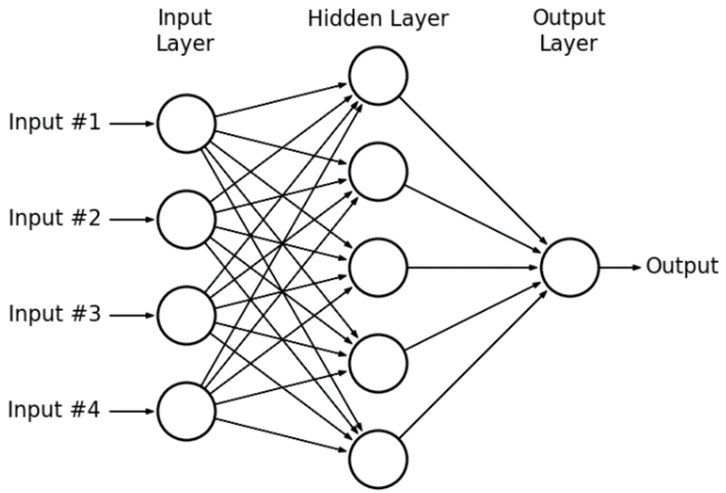
- Perceptron with just feature f_1 cannot separate the data
- Could we add another feature to our perceptron $f_2 = f_1^2$
  - Note could also think of this as just using feature f_1 but now allowing a quadric surface to separate the data
- Could we solve the problem without adding the input dimensions?

Multi-Layer Perceptron Learning?

- Minsky & Papert (1969) offered solution to XOR problem by combining perceptron unit responses using a second layer of Units.
- Piecewise linear classification using an MLP with threshold (perceptron) units



Multilayer Perceptron (MLP)



- Stacking multiple layers of perceptrons (adding hidden layers) makes a multilayer perceptron (MLP)
- MLP can engage in sophisticated decision making, where single-layer perceptrons fail, e.g. XOR problem.
- Try it: http://playground.tensorflow.org
- Multi-layer neural networks will allow us to discover high-order features automatically from the input space